



Girijananda Chowdhury University (GCU)-Assam
Department of Civil Engineering
Surveying Engineering Laboratory
MCQ Questions and Answers

Key Surveying Concepts

- **Fundamental Principle:** Surveying aims to work from the whole to the part to prevent error accumulation.
- **Chain Surveying:** Gunter's chain is 66 ft long with 100 links. Key processes include ranging (aligning intermediate points) and establishing the base line.
- **Levelling & Contouring:** Levelling is used for determining elevations, while cross-sectioning is often best for hilly terrain.
- **Instruments:** EDM stands for Electronic Distance Measurement (used in total stations), and theodolites are used for measuring horizontal angles.
- **Survey Types:** Curvature of the earth is ignored in plane surveying, as opposed to geodetic surveying.
- **Objective:** The primary goal is creating accurate maps or plans.

Surveying MCQ (Multiple Choice Questions)

1. What is surveying?
a) Surveying is used to find the elevations of given points with respect to given or assumed datum
b) Surveying shows the relative positions of the objects on the surface of the earth
c) Surveying is to find the elevation of points having the same contour interval
d) All of the mentioned Answer: b
Explanation: Surveying is defined as determining the relative positions of points above or beneath the surface of the earth by means of direct or indirect measurements of distance and direction and elevation.
2. Which of the following is the first principle of surveying?
a) Whole to whole
b) Part to part
c) Part to whole
d) Whole to part Answer: d
Explanation: The first principle of surveying is to work from whole to part. Before starting the actual survey measurements, the surveying is to work from around the area to fix the best positions of survey lines and survey stations.
3. Which of the following type of surveying is used for exploring mineral wealth?
a) Military surveying
b) Mine surveying
c) Topographic surveying
d) Engineering surveying Answer: b
Explanation: For exploring mineral wealth mine surveying is used. Determining points of strategic importance is military surveying.
4. In which of the following type of surveying only linear measurements are made?
a) Dumpy level
b) Theodolite surveying
c) Chain surveying
d) Contouring Answer: c
Explanation: Chain surveying is the type of surveying in which only linear measurements are made in the field. This type of surveying is suitable for surveys of small extent on open take simple details.
5. Which of the following classification in surveying is based on the instrument used?
a) Traverse surveying
b) Cadastral surveying
c) Topographic surveying
d) Hydrographic surveying Answer: a
Explanation: Topographic surveying, Hydrographic surveying, Cadastral surveying classification is based on the nature of field survey. Traverse surveying, chain surveying is classified based on the type of instrument used.
6. In which of the following areas does compass surveying is not recommended?
a) Large areas
b) Undulating areas
c) Crowded with many details



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d) Local attraction suspected areas Answer: d

Explanation: Compass surveying is not recommended for areas where the local attraction is suspected due to the presence of a magnetic substance.

7. In which of the following cases compass surveying is recommended?

- a) When area is small, undulating and not details are crowded
- b) When area is large, undulating and crowded with many details
- c) When area is small, even and crowded with many details
- d) When area is large, even and crowded with many details

Answer: b

Explanation: Compass surveying is recommended when the area is large, undulating and crowded with many details. It is not recommended for areas where the local attraction is suspected due to the presence of a magnetic substance.

8. Which of the following is not required for chain surveying?

- a) Dumpy level
- b) Pegs
- c) Arrows
- d) 20 m chain

Answer: a

Explanation: Dumpy level is used in levelling not in chain surveying. 20 m chain, arrows, pegs etc are mandatory for chain surveying.

9. Which of the following is the last step in chain surveying?

- a) Fixing
- b) Reconnaissance
- c) Running survey lines
- d) Marking

Answer: c

Explanation: After having completed the preliminary work, the chaining may be started from the base line. The work in running a survey line is twofold, to chain the line and to locate the adjacent details.

10. Which of the following cannot be done with the help of theodolite in surveying?

- a) Measuring horizontal distances
- b) Prolonging survey lines
- c) Laying off horizontal angles
- d) Locating points on lines

Answer: a

Explanation: Theodolite is the most precise instrument designed for the measurement of horizontal and vertical angles. It has wide applicability in surveying such as laying off horizontal angles, locating points on line, prolonging survey lines, establishing grades etc.

11. Which of the following is an indirect method of surveying?

- a) Countouring
- b) Chain surveying
- c) Tacheometry
- d) All of the mentioned

Answer: c

Explanation: Generally, horizontal distances are measured by direct methods, i.e. laying of chains or tapes on the ground. These methods are not always convenient if the ground is undulating, rough, difficult and inaccessible. Under these circumstances, indirect methods are used to obtain distances. One such method is "Tacheometry".

12. Which of the following branch of surveying is used to find the elevations of given points with respect to given or assumed datum?

- a) Plane table surveying
- b) Traversing
- c) Contouring
- d) Levelling

Answer: d

Explanation: Levelling is a branch of surveying is used to find the elevations of given points with respect to given or assumed datum.

13. Which of the following are the principles of surveying?

- a) Covering the entire area
- b) Working from whole to part
- c) Taking measurements
- d) Determining the elevation differences

Answer: b



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Explanation: By working from whole to part, it is possible to eliminate the errors and to localise the errors. Otherwise, it might expand in magnitude.

14. Which of the following surveying methods is meant to be having high precision?

- a) Terrestrial photogrammetry
- b) Traverse surveying
- c) Aerial photogrammetry
- d) Theodolite surveying

Answer: c

Explanation: Though terrestrial photogrammetry is having accuracy in the obtained values, aerial photogrammetry is capable of producing precise output when compared to the remaining methods. This accuracy makes it different from the remaining methods and is recommended when high quality works are conducted.

15. Which of the following doesn't describe the use of hydrographic surveying?

- a) Nautical charts for navigation
- b) Establishing mean sea level
- c) Laying an Alignment
- d) Making underground investigations

Answer: b

Explanation: Hydrographic surveying can find its use in making nautical charts for navigation, making underground investigations for construction, establishing mean sea level, determining shore line etc.

16. Which of the following type of ranging is done if both ends of surveying lines are visible?

- a) Indirect
- b) Reciprocal
- c) Unable to do
- d) Direct

Answer: d

Explanation: Direct ranging is done if both ends of surveying lines are visible. Indirect ranging is done when both ends are not intervisible.

17. Which of the following is not a method of levelling?

- a) Spirit levelling
- b) Traverse levelling
- c) Barometric levelling
- d) Trigonometric levelling

Answer: b

Explanation: Three principal methods are used for determining a difference in elevation, namely, barometric levelling, trigonometric levelling and spirit levelling. Traversing is that type of surveying in which a number of connected survey lines form the framework.

18. In which of the following type of surveying in the mean surface of the earth is considered as a plane and the spheroidal shape is neglected?

- a) Plane Surveying
- b) Geodetic Surveying
- c) Hydrographic Surveying
- d) Topographic Surveying

Answer: a

Explanation: The type of surveying in which the mean surface of the earth is considered as a plane and the spheroidal shape is neglected is plane surveying.

19. Which of the following doesn't involve the method of traversing?

- a) Plane Table surveying
- b) Tacheometric surveying
- c) Chain surveying
- d) Theodolite surveying

Answer: b

Explanation: Depending on the instruments used in determining the relative directions of the traverse lines. The principal methods adopted are Chain traversing, Compass traversing, Transit tape traversing, Plane-table traversing. Tacheometric surveying involves a lot of instrumental work rather than ground work.

20. Which of the following is not a method of plane table surveying?

- a) Trisection
- b) Intersection
- c) Resection
- d) Radiation

Answer: a



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Explanation: The methods which are adopted in case of plane table surveying involve radiation, Intersection, resection and traversing, which are used based on the type of output required.

21. While taking Observations for the height and distances, which of the following method of surveying is used?

- a) Plane surveying
 - b) Geodetic surveying
 - c) Chain surveying
 - d) Compass surveying
- Answer: b

Explanation: Geodetic surveying is used because it is assumed that the distances between the points observed are not large so that either the effect of curvature and refraction may be neglected or proper corrections may be applied linearly.

22. Which of the following type of surveying can be employed in a magnetic area?

- a) Compass surveying
 - b) Traverse surveying
 - c) Plane table surveying
 - d) Theodolite surveying
- Answer: c

Explanation: Other than plane table surveying, remaining methods need a compass for initiating the work in which it is not possible to access compass in a magnetic area. Plane table surveying can be employed there as it can be done without usage of compass.

23. Which of the following survey deals with bodies of water for the purpose of navigation, water supply, harbor works or for the determination of mean sea level?

- a) City surveying
 - b) Cadastral surveying
 - c) Topographic surveying
 - d) Hydrographic surveying
- Answer: d

Explanation: Survey which deals with bodies of water for the purpose of navigation, water supply, harbor works or for the determination of mean sea level is hydrographic surveying.

24. Determining points of strategic importance are called _____

- a) Traverse surveying
 - b) Military surveying
 - c) City surveying
 - d) Topographic surveying
- Answer: b

Explanation: Determining points of strategic importance is military surveying. City surveying is made in connection with the construction of streets, water supply systems and sewers.

25. Which of the following is not a natural error in compass surveying?

- a) Local attraction due to the proximity of local attraction forces
 - b) Pivot being bent
 - c) Magnetic changes in the atmosphere due to clouds and Strom's
 - d) Variation in declination
- Answer: b

Explanation: Instrumental errors are those which arise due to the faulty adjustment of the instrument e.g pivot being bent, blunt pivot point, etc. Variation in declination, magnetic changes in the atmosphere due to clouds and Strom's etc also comes under natural errors in compass surveying.

26. Which of the following processes has more accuracy in its output?

- a) Plane table surveying
 - b) Chain surveying
 - c) Compass surveying
 - d) Total station
- Answer: d

Explanation: Plane table surveying is a rough process conducted in order to have an idea of the land and its condition for construction. Due to this, it is not capable of producing accurate output in its recording.

27. Marine surveying is a necessary in case of boat designing.

- a) False
 - b) True
- Answer: b

Explanation: Marine survey involves certain features which are capable of determining the boat functioning and its capabilities. The main purpose of marine surveying includes the processing of an errorless functioning boat, which can be used while underground surveying.



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28. Which of the following will not come under the marine surveying category?

- a) Water survey
- b) Yacht survey
- c) Machinery survey
- d) Cargo survey

Answer: a

Explanation: Marine surveying involves certain classifications which can be given as yacht survey, machinery survey and cargo survey. All these are under the roof of marine surveying, which done to improve the benefit of the instrument or object used.

29. Which of the following is having the same principle as that of photo-theodolite?

- a) Theodolite surveying
- b) Plane-table surveying
- c) Traverse surveying
- d) Compass surveying

Answer: b

Explanation: Photo-theodolite involves the direction of same objects in the photograph by which base is measured is known and positions are located with extremities. It is the same as that of plane table surveying which indicates both are having same principle.

30. In which of the following cases the method of deflection distances is used?

- a) Land surveys
- b) Road survey
- c) Town survey
- d) Railway planning survey

Answer: b

Explanation: The deflection distances method is having at most priority in case of road surveys as the curvature for joining parallel straights is to be done without any error.

Chain Surveying – Chain Triangulation

31. What is the lateral distance of an object or ground feature measured from a survey line?

- a) Offset
- b) Perpendicular distance
- c) Side distance
- d) Perpendicular offset

Answer: a

Explanation: An offset is the lateral distance of an object or ground feature measured from a survey line. When the angle of the offset is 90° , it is called perpendicular offset.

32. An offset is laid out 6° from its true direction on the field. Find the resulting displacement of the plotted point on the paper in a direction parallel to the chain line? (Given the length of offset is 10 m and scale is 5 m to 1 cm)

- a) 0.209 cm
- b) 0.260 cm
- c) 0.0109 cm
- d) 0.910 cm

Answer: a

Explanation: Distance parallel to the chain = $l \sin \Delta / s = 10 \sin 6^\circ / 5 = 0.209$ cm.

33. An offset is laid out 6° from its true direction on the field. Find the resulting displacement of the plotted point on the paper in a direction perpendicular to the chain line? (Given length of offset is 10 m and scale is 5 m to 1 cm)

- a) 0.209 cm
- b) 0.260 cm
- c) 0.0109 cm
- d) 0.910 cm

Answer: c

Explanation: Distance perpendicular to the chain = $l(1 - \cos \Delta) / s = 10(1 - \cos 6^\circ) / 5$. By simplifying we get 0.0109 cm.

34. An offset is laid out $1^\circ 30'$ from its true direction on the field. Find the degree of accuracy with which the offset should be measured so that the maximum displacement of the point on the paper from both sources may be equal?

- a) 1 in 38
- b) 1 in 39
- c) 1 in 36
- d) 1 in 37

Answer: b

Explanation: Displacement due to angular error is $l \sin \Delta$. Displacement due to linear error l/r . Equating both $r = 38.20$. Hence, the offset should be measured with an accuracy of 1 in 39.



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35. An offset is measured with an accuracy of 1 in 40. If the scale of plotting is 1 cm = 20 m, find the limiting length of the offset so that the displacement of the point on the paper from both sources of error may not exceed 0.25 mm?

- a) 14.10 m
- b) 14.14 m
- c) 14.40 m
- d) 14.44 m

Answer: b

Explanation: The total displacement of the paper = $\sqrt{2} l / \text{rs.}$ From this we can simply for $l = (0.025/\sqrt{2}) * 40 * 20 = 14.14 \text{ m.}$

36. What triangles are generally preferred to get good results in plotting?

- a) Isosceles
- b) Obtuse angled
- c) Equilateral
- d) Acute angle

Answer: c

Explanation: To get good results in plotting, the frame work should consist of triangles which are as nearly equilateral as possible.

37. The line must run through level ground as possible is the condition to be fulfilled by survey lines or survey stations.

- a) True
- b) False

Answer: a

Explanation: The survey stations should be so selected that a good system of lines is obtained fulfilling the following conditions such as survey stations must be mutually visible, the lines must run through level ground as possible etc.

38. The main lines should form well-conditioned triangles is the condition to be fulfilled by survey lines or survey stations.

- a) True
- b) False

Answer: a

Explanation: The survey stations should be so selected that a good system of lines is obtained fulfilling the following conditions such as the lines must run through the level ground as possible, the main lines should form well-conditioned triangles etc.

39. Survey lines must be as few as possible is the condition to be fulfilled by survey lines or survey stations.

- a) True
- b) False

Answer: a

Explanation: The survey stations should be so selected that a good system of lines is obtained fulfilling the following conditions such as the lines must run through the level ground as possible, survey lines must be as few as possible so that the frame work can be plotted conveniently, the main lines should form well-conditioned triangles etc.

40. As far as possible main survey lines should pass through the obstacles to get know how many obstacles present.

- a) True
- b) False

Answer: b

Explanation: The survey stations should be so selected that a good system of lines is obtained fulfilling the following conditions such as the lines must run through the level ground as possible, as far as possible, the main survey lines should not pass through obstacles.

41. How many kinds of obstacles of chaining are there?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: Obstacles of chaining are of three kinds. They are obstacles to ranging, obstacles to chaining, obstacles to both chaining and ranging.

42. Which of the following is not one among the three major kinds of obstacles of chaining?

- a) Obstacles to ranging
- b) obstacles to chaining
- c) obstacles to levelling
- d) obstacles to ranging and chaining

Answer: c

Explanation: Obstacles to levelling is not a kind of obstacles to chaining. Obstacles of chaining are of three kinds. They are obstacles to ranging, obstacles to chaining, obstacles to both chaining and ranging.

43. When it is not possible to chain round the obstacle e.g. a river. This case comes under which among the three kinds of obstacles to chaining?

- a) obstacles to ranging but not chaining



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- b) obstacles to chaining but not ranging
- c) obstacles to levelling
- d) obstacles to ranging and chaining

Answer: b

Explanation: There may be two cases of this obstacle i.e obstacle to chaining but not ranging, when it is possible to chain round the obstacle, i.e a pond, hedge etc and when it is not possible to chain round the obstacle e.g. a river.

44. To continue a survey line AB past an obstacle, a line BC 100 m long was set out perpendicular to AB and from C angles BCD and BCE were set out at 60° and 45° respectively. Determine the lengths which must be chained off along CD in order that ED may be in AB produced?
- a) 100 m
 - b) 200 m
 - c) 300 m
 - d) 400 m

Answer: b

Explanation: Here angle ABC is 90° . From, ΔBCD , $CD = BC \sec 60^\circ = 100 \times 2 = 200\text{m}$.

45. If we select two points A and B on either side of the obstacle and equal perpendiculars AC and BD are set out. Then AB is equal to?
- a) AC
 - b) CD
 - c) DA
 - d) BD

Answer: b

Explanation: Since AC and BD are two perpendiculars set either side of the obstacle of equal length. Therefore CD is parallel and equal to AB. Therefore, $AB = CD$.

Surveying Questions and Answers – Transit Theodolite

46. _____ is the most precise instrument designed for the measurement of horizontal and vertical angles.
- a) Survey chain
 - b) Dumpy level
 - c) Theodolite
 - d) Telescope

Answer: c

Explanation: Theodolite is the most precise instrument designed for the measurement of horizontal and vertical angles. It has wide applicability in surveying such as laying off horizontal angles, locating points on line, prolonging survey lines, establishing grades etc.

47. Which of the following cannot be done with the help of theodolite in surveying?
- a) Laying off horizontal angles
 - b) Locating points on lines
 - c) Prolonging survey lines
 - d) Measuring horizontal distances

Answer: d

Explanation: Theodolite is the most precise instrument designed for the measurement of horizontal and vertical angles. It has wide applicability in surveying such as laying off horizontal angles, locating points on line, prolonging survey lines, establishing grades etc.

48. Which of the following cannot be done with the help of theodolite in surveying?
- a) Establishing grades
 - b) determining the difference in elevation
 - c) Setting out curves
 - d) determining the area of ground

Answer: d

Explanation: Theodolite is the most precise instrument designed for the measurement of horizontal and vertical angles. It has wide applicability in surveying such as laying off horizontal angles, locating points on line, prolonging survey lines, establishing grades, determining the difference in elevation, setting out curves etc.

49. A transit theodolite is one in which the line of sight can be reversed by revolving the telescope through 180° in a vertical plane.
- a) True
 - b) False

Answer: a

Explanation: Theodolite may be classified as transit and non-transit theodolite. A transit theodolite is one in which the line of sight can be reversed by revolving the telescope through 180° in a vertical plane.



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50. A non-transit theodolite is one in which the line of sight can be reversed by resolving the telescope through 180° in a vertical plane.

a) True

b) False Answer: b

Explanation: Theodolite may be classified as transit and non-transit theodolite. The non-transit theodolites are those in which the telescope cannot be transited.

51. The transit theodolites are also called plain theodolites.

a) True

b) False Answer: b

Explanation: The non-transit theodolites are also called plain theodolites. In which the telescope cannot be transited.

52. How many types do theodolites classified?

a) 2

b) 3

c) 4

d) 5

Answer: a

Explanation: Theodolites are classified into two types. They are transit theodolites and non – transit theodolites.

53. Which of the following is an integral part of the theodolite and is mounted on a spindle known as a horizontal axis?

a) Telescope

b) Index frame

c) Horizontal plane Vernier

d) Horizontal circle

Answer: a

Explanation: The telescope is an integral part of the theodolite. It is mounted on a spindle known as a horizontal axis or trunnion axis.

54. Horizontal axis is also called _____

a) Inner axis

b) Outer axis

c) Trunnion axis

d) Line of sight

Answer: c

Explanation: The telescope is an integral part of the theodolite. It is mounted on a spindle known as a horizontal axis or trunnion axis.

55. The vertical circle is a circular graduated arc attached to the _____ axis of the telescope.

a) inner axis

b) outer axis

c) trunnion axis

d) line of sight

Answer: c

Explanation: The vertical circle is a circular graduated arc attached to the telescope. Consequently, the graduated arc rotates with the telescope when the latter is turned about the horizontal axis.

56. By means of vertical circle clamp and its corresponding _____ the telescope can be set accurately at any desired position in the vertical plane.

a) tripod head

b) focusing screw

c) levelling head

d) tangent screw

Answer: d

Explanation: By means of vertical circle clamp and its corresponding slow motion or tangent screw the telescope can be set accurately at any desired position in the vertical plane.

57. The index frame is _____ shaped frame.

a) U

b) V

c) T

d) A

Answer: c

Explanation: The index frame is a T shaped frame consisting of a vertical leg known as clipping arm. It is also called a Vernier frame.

58. In theodolites, two standards resemble the letter _____

a) U



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- b) V
- c) T
- d) A

Answer: d

Explanation: Two standards resemble letter A are mounted on the upper plates. The trunnion axis of the telescope is supported on these.

59. Which of the following is not a function of levelling head?

- a) To support the main part of the instrument
- b) To attach the theodolite to the tripod
- c) To provide a mean for levelling the theodolite
- d) To provide the exact centering over the station mark

Answer: d

Explanation: A levelling head has three distinctive functions. They are to support the main part of the instrument, to attach the theodolite to the tripod and to provide a mean for levelling the theodolite.

60. The plate level can be centered with the help of _____

- a) Focusing screw
- b) Foot screw
- c) Tangent screw
- d) Clip screw

Answer: b

Explanation: The plate level can be centered with the help of foot screw. In some theodolites only one plate level is provided.

61. On unclamping the upper screw and clamping the lower clamp, the instrument can rotate on its inner axis with relative motion between vernier and scale.

- a) True
- b) False

Answer: a

Explanation: The lower clamp is clamped and the upper clamp is unclamped, the upper plate and the instrument can rotate on the inner axis with a relative motion between the vernier and the scale.

62. To measure the horizontal angle which of the following is the first step?

- a) Releasing all clamps
- b) Levelling instrument
- c) Turning plates
- d) Clamping the plates

Answer: b

Explanation: To measure the horizontal angle, firstly we set up instrument and level it. Secondly, releases all clamps. Turning the upper and lower plates in opposite directions till the zero of the one of the vernier is against the zero of the scale. Next clamp the plates together by upper clamp and lower clamp and bring the two zeros into exact coincidence by turning upper tangent screw. It continues next procedure.

63. To measure a horizontal angle, after releasing all clamps, we turn the upper and lower plates in the same direction.

- a) True
- b) False

Answer: b

Explanation: After releasing all clamps. Turning the upper and lower plates in opposite directions till the zero of the one of the vernier is against the zero of the scale.

64. If one of the vernier is at 0° then another vernier reading shows / also shows _____

- a) 90°
- b) 0°
- c) 180°
- d) 45°

Answer: c

Explanation: Turning the upper and lower plates in opposite directions till the zero of the one of the vernier is against the zero of the scale. Next clamp the plates together by upper clamp and lower clamp and bring the two zeros into exact coincidence by turning upper tangent screw. Now when we take reading on another vernier it should be 180° , if there is no instrument error.

65. To measure a horizontal angle by direct method is also called a repetition method.

- a) True
- b) False

Answer: b

Explanation: To measure a horizontal angle by direct method is also called the reiteration method. It is suitable for measurements of angles of a group having a common vertex point.



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66. In measuring direct angles, azimuths from the back line may vary from 0° to 180° .

- a) True
- b) False

Answer: b

Explanation: Direct angles are angles measured clockwise from the preceding line to the following line. They are also known as angles to the right or azimuths from the back line. They may vary from 0° to 360° .

67. A deflection angle is an angle in which a survey line makes with the prolongation of the preceding line.

- a) True
- b) False

Answer: a

Explanation: A deflection angle is an angle in which a survey line makes with the prolongation of the preceding line. It is designated as right (R) or left (L) according to its measure to the clockwise or to anti clockwise from the prolongation of the previous line.

68. To run a straight line between two points, when both ends are inter visible. We establish intermediate points through

- a) line of sight
- b) balancing
- c) using random line
- d) back sight

Answer: a

Explanation: Set instrument at A and take sight on B. Establish intermediate points C, D, E etc. in the line of sight. It is possible only in case both ends are inter visible.

69. Sources of errors in transit work are broadly classified into _____ types.

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: Sources of error in theodolite work are 3 types. They are instrumental, personal and natural.

70. Error due to imperfect adjustment of plate levels comes under _____ error.

- a) personal
- b) natural
- c) instrumental
- d) personal and natural

Answer: c

Explanation: Instrumental errors are due to the imperfect adjustment of an instrument, structural defects in the instrument and imperfections due to wear.

71. Error due to in accurate centering comes under _____ error.

- a) personal
- b) natural
- c) instrumental
- d) personal and natural

Answer: a

Explanation: The personal errors may be due to errors in manipulation, errors in sighting and reading. Inaccurate centering comes under errors in manipulation.

72. Parallax comes under _____ error.

- a) personal
- b) natural
- c) instrumental
- d) personal and natural

Answer: a

Explanation: The personal errors may be due to errors in manipulation, errors in sighting and reading. Parallax comes under errors due to sighting and reading.

73. Unequal settlement of tripod comes under _____ source of error.

- a) personal
- b) natural
- c) instrumental
- d) personal and natural

Answer: b

Explanation: Unequal atmospheric refraction due to high temperature comes under natural errors. Unequal settlement of tripod, wind vibrations etc., also comes under the same category.



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74. What is the term used for an imaginary line on the ground joining points of equal elevation?

- a) Level line
- b) Line of sight
- c) Datum
- d) Contour

Answer: d

Explanation: A Contour is an imaginary line on the ground joining the points of equal elevation. It is a line in which the surface of the ground is interested in a level surface.

75. Which of the following indicates the elevations directly?

- a) Level line
- b) Line of sight
- c) Datum
- d) Contour

Answer: d

Explanation: A Contour is an imaginary line on the ground joining the points of equal elevation. It is a line in which the surface of the ground is interested by a level surface. On a plan, the relative altitudes of the points can be represented by shading, Hachures, form lines or contour lines. Out of these, contour lines are most widely used because they indicate the elevations directly.

Surveying Questions and Answers – Prismatic Compass

76. Which of the following is the most convenient and portable instrument for direct measurement of directions?

- a) Prismatic compass
- b) Surveyor's compass
- c) Theodolite
- d) Sextant

Answer: a

Explanation: Prismatic compass is the most convenient and portable form of a magnetic compass which can either be used as a hand instrument or can be fitted on a tripod.

77. Which of the following is not a part of the prismatic compass?

- a) Agate cap
- b) Prism cap
- c) Brake pin
- d) Jewel bearing

Answer: d

Explanation: Prism cap, prism, brake pin, spring brake, pivot, agate cap etc are parts of the prismatic compass. Jewel bearing is one of the parts of surveyor's compass.

78. In which of the following compass needle does not act as an index?

- a) Prismatic compass
- b) Surveyor's compass
- c) Theodolite
- d) Sextant

Answer: a

Explanation: In the case of prismatic compass needle does not act as an index. In the case of surveyors, compass needle acts as an index.

79. In prismatic compass, graduations are in W.C.B system.

- a) True
- b) False

Answer: a

Explanation: In prismatic compass, graduations are in W.C.B system, having 0° to south end, 90° at west, 180° at north and 270° at east. The graduations are engraved inverted.

80. In which of the following compass sighting and reading taking can be done simultaneously from one position of the observer?

- a) Prismatic compass
- b) Surveyor's compass
- c) Theodolite
- d) Sextant

Answer: a

Explanation: In case of a prismatic compass, reading is taken with the help of prism provided at the eye slit. Sighting and reading taking can be done simultaneously from one position of the observer.

81. In surveyor's compass, graduations are in a Q.B system.

- a) True



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b) False Answer: a

Explanation: In Surveyor's compass, the graduations are in a Q.B system, having 0° at north and south and 90° at east and west. East and West are interchanged.

Surveying Questions and Answers – Setting out Simple Curves

82. Which among the following indicates the correct set of methods for setting out a simple curve?

- a) Angular method, curvature method
- b) Linear method, angular method
- c) Curvature method, linear method
- d) Tangent method, curvature method

Answer: b

Explanation: The methods involved in setting out a curve can be given as linear method, angular method and instrument method is also involved which is capable of providing accurate information.

83. In linear method of setting out curve, which of the following is not used?

- a) Tape
- b) Chain
- c) Theodolite
- d) Compass

Answer: c

Explanation: Theodolite is an instrument that is used for computing angles or bearings between the lines which are used for further calculation in computing the distance between the points. Whereas tape, chain can be directly used for determining the distance.

84. Which of the following methods is used when curve to be designed is short?

- a) Linear method
- b) Angular method
- c) Tangent method
- d) Curvature method

Answer: a

Explanation: Linear method adopts the usage of chain or tape in which only distance can be found but not angles. Moreover, this process is used when accuracy is not required and speed of work is concerned.

85. For setting the tangent, which process is most commonly used?

- a) Rankine's method
- b) Trial and error method
- c) Tacheometric method
- d) Two theodolite method

Answer: b

Explanation: Though trial and error method is a tedious process, it is mostly used in all methods of setting a tangent because it involves accuracy and might be able to deliver the best output through calculation.

86. In angular method of setting a curve, which of the following is used?

- a) Compass
- b) Tape
- c) Chain
- d) Theodolite

Answer: d

Explanation: The angular method of setting out a curve involves finding out the angles between the lines, which is required for further assessment. Before measuring angles it is essential to set the required asserts.

87. For setting the tangent, which process is most commonly used?

- a) Rankine's method
- b) Trial and error method
- c) Tacheometric method
- d) Two theodolite method

Answer: b

Explanation: Though trial and error method is a tedious process, it is mostly used in all methods of setting a tangent because it involves accuracy and might be able to deliver the best output through calculation.

88. P.T, P.I, P.C are the basic requirements for setting a curve.

- a) True
- b) False

Answer: a

Explanation: The point of curvature, point of tangency, point of intersection are basic requirements which are used for setting out curve, tangents by placing instrument at that point based on our requirement.



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89. Find the value of radius if the value of D is given as 23.76m.

- a) 214.98m
- b) 241.61m
- c) 214.16m
- d) 241.16m

Answer: d

Explanation: The value of radius with respect to only radius can be given as,

$R = 5730/D$. On substitution, we get

$$R = 5730 / 23.76$$

$$R = 241.16 \text{ m.}$$

90. Find the value of length of the curve if the degree of curve is taken at 20m arc with an angle $42^\circ 12'$.

- a) 24.9m
- b) 24.2m
- c) 42.2 m
- d) 49.2m

Answer: c

Explanation: The value of the length of the curve at an arc which is designated at 20 m can be given as, $l = 20 \cdot \theta / D$. On substitution, we get

$$l = 20 \cdot 42^\circ 12' / 20$$

$$l = 42.2 \text{ m.}$$

91. Using the degree of curvature, find the value of radius of curve if the distance is given as 24.65 m.

- a) 64.49m
- b) 46.49m
- c) 46.94m
- d) 64.94m

Answer: b

Explanation: In degree of curvature, the value of radius can be designated by using 20m arc length. The formula is given as

$R = 1146 / D$. On substitution, we get

$$R = 1146 / 24.65 = 46.49 \text{ m.}$$

Surveying Questions and Answers – EDM – Total Station

92. In total station, data is stored in _____

- a) Pen drive
- b) Data card
- c) Micro processor
- d) External hardware

Answer: c

Explanation: Micro processor provided in the instrument helps in saving the data and helps in transferring it to the system, by which it can be viewed in different software.

93. Vertical angle is measured in the total station as Zenith angle.

- a) False
- b) True

Answer: b

Explanation: The vertical angle is usually measured as a zenith angle which indicates 0° vertically up, 90° at horizontal, and 180° vertically down. The zenith angle is generally easier to work.

94. Which of the following indicates the formula for converting slope distance to horizontal distance?

- a) $S = H (\sin z)$
- b) $H = S \cdot S (\sin z)$
- c) $H \cdot H = S (\sin z)$
- d) $H = S (\sin z)$

Answer: d

Explanation: The slope distance obtained can be used for calculating horizontal distance from the formula, $H = S (\sin z)$ where, H= horizontal distance, S = slope distance.

95. When total station is sighted to the target, which of the operation acts first?

- a) Rotation of optical axis
- b) Rotation of vertical axis
- c) Rotation of horizontal axis
- d) Rotation of line of collimation

Answer: a

Explanation: At the time of sighting the instrument towards the target, first step involves the rotation of the instrument's optical axis from the instrument north in horizontal plane.